

QUEEN ALIA INTL, Jordan

WMO: 402720

Lat: 31.723N

Lon: 35.993E

Elev: 2395

StdP: 13.47

Time Zone: 2.00 (JOR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.6	33.6	11.8	11.0	51.7	16.3	13.7	53.2	33.2	46.1	29.2	44.8	2.7	40	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.0	98.8	67.0	96.4	66.1	93.5	65.7	72.8	89.8	71.1	88.4	69.5	86.5	11.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.8	111.9	79.5	65.9	104.5	79.0	64.1	98.0	77.8	38.2	89.3	36.5	88.6	35.1	86.4	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	22.0	19.5	DB	26.4	104.5	3.4	2.7	24.0	106.5	22.0	108.0	20.1	109.5	17.7	111.5
			WB	24.5	75.5	3.9	4.3	21.7	78.6	19.4	81.1	17.2	83.5	14.4	86.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.8	46.4	48.7	53.8	60.9	68.9	73.5	76.8	76.9	73.5	67.3	57.2	49.2	(6)
(7)		DBStd	12.26	4.53	5.47	6.79	7.37	6.53	4.38	3.63	3.65	4.27	5.76	5.90	5.04	(7)
(8)		HDD50	329	130	81	32	3	0	0	0	0	0	9	74	(8)	
(9)		HDD65	2363	578	458	356	168	32	0	0	0	39	242	490	(9)	
(10)		CDD50	5014	17	44	149	331	587	704	832	833	704	538	225	50	(10)
(11)		CDD65	1573	0	0	8	46	154	255	367	368	255	111	9	0	(11)
(12)		CDH74	22390	0	11	171	833	2496	3834	5127	4960	3299	1497	153	9	(12)
(13)		CDH80	11097	0	1	45	289	1145	1946	2809	2705	1573	557	26	1	(13)
(14)	Wind	WSAvg	7.7	7.9	8.4	8.3	8.1	8.2	8.5	9.1	7.8	6.5	5.7	6.8	7.1	(14)
(15)	Precipitation	PrecAvg	8.7	2.3	2.1	1.2	0.4	0.1	0.0	0.0	0.0	0.0	0.2	0.9	1.6	(15)
(16)		PrecMax	15.8	4.5	7.5	3.4	2.2	0.8	0.0	0.0	0.0	0.0	0.9	4.0	5.4	(16)
(17)		PrecMin	3.4	0.1	0.5	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.7	1.0	1.4	0.9	0.5	0.2	0.0	0.0	0.0	0.0	0.2	0.9	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.7	75.3	85.6	92.0	98.7	100.5	103.8	102.3	100.0	93.6	83.7	74.8	(19)
(20)		MCWB	51.6	52.9	57.6	60.2	64.8	65.2	69.3	69.2	66.6	62.7	57.5	52.2	(20)	
(21)		2%	DB	64.2	69.7	78.7	87.4	94.5	96.9	98.5	97.2	95.1	89.8	78.6	68.1	(21)
(22)		MCWB	49.1	51.0	55.1	58.3	62.9	64.2	69.2	68.0	65.1	62.4	56.1	51.1	(22)	
(23)		5%	DB	60.4	64.6	73.5	82.8	90.7	93.3	94.9	94.8	91.7	86.2	73.8	63.3	(23)
(24)		MCWB	48.2	49.2	53.4	56.7	61.7	64.3	66.8	67.7	65.3	62.0	55.1	49.6	(24)	
(25)		10%	DB	56.9	60.6	68.4	78.7	86.3	89.9	91.9	91.8	88.4	82.4	70.1	60.2	(25)
(26)		MCWB	47.3	48.0	51.3	55.3	60.6	63.9	66.3	67.0	65.2	61.0	54.2	48.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	56.0	60.0	63.3	70.3	71.1	75.8	75.5	73.8	70.6	62.1	56.4	(27)
(28)		MCDWB	62.6	71.0	77.7	85.4	91.4	89.9	95.3	91.7	86.4	84.9	72.1	66.5	(28)	
(29)		2%	WB	51.5	52.7	57.2	60.6	66.1	68.9	72.4	73.0	70.9	66.9	59.2	53.5	(29)
(30)		MCDWB	59.7	64.2	73.6	79.5	87.8	88.2	90.6	90.0	85.8	80.2	70.9	62.0	(30)	
(31)		5%	WB	49.6	50.8	54.9	58.5	63.6	67.3	70.3	71.3	69.0	64.7	57.5	51.8	(31)
(32)		MCDWB	57.3	61.1	69.5	77.9	84.9	86.8	88.6	88.6	84.0	77.7	69.0	60.2	(32)	
(33)		10%	WB	48.1	49.1	52.8	56.5	61.5	65.6	68.7	69.7	67.5	63.0	55.8	50.2	(33)
(34)		MCDWB	55.1	58.2	65.7	75.4	82.1	85.1	86.5	86.1	82.6	77.1	67.5	57.4	(34)	
(35)	Mean Daily Temperature Range		MDBR	19.5	20.9	24.3	28.5	30.7	31.8	31.0	31.0	30.6	28.4	24.5	21.2	(35)
(36)		5% DB	MCDBR	26.4	28.3	31.7	33.7	35.1	36.7	34.8	34.1	34.5	32.1	28.9	26.9	(36)
(37)		MCWBR	14.8	14.3	15.2	15.0	15.6	16.3	15.3	14.1	14.3	14.9	15.1	15.0	(37)	
(38)			MCDBR	22.5	24.1	28.2	30.8	32.4	33.1	31.8	31.5	30.9	28.0	24.7	22.2	(38)
(39)		5% WB	MCWBR	14.2	13.8	15.0	15.2	16.7	16.6	15.8	15.0	14.6	14.9	14.4	14.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.345	0.376	0.422	0.469	0.468	0.394	0.400	0.407	0.403	0.422	0.369	0.347	(40)	
(41)		taud	2.254	2.153	2.005	1.888	1.918	2.161	2.154	2.138	2.146	2.076	2.208	2.257	(41)	
(42)		Ebn at Noon	274	276	271	263	263	282	280	277	273	256	262	266	(42)	
(43)		Edh at Noon	35	43	54	63	62	48	48	48	46	46	37	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1004	1263	1703	2064	2387	2646	2567	2356	2018	1546	1160	941	(44)	
(45)		RadStd	57	87	91	83	95	35	31	32	49	57	56	54	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.86	N/A	-4.52	N/A	N/A	N/A	N/A	N/A	+288	+193
(47)	Regional (18 neighbors)	+0.79	N/A	-3.51	N/A	N/A	N/A	N/A	-108	+252	+166

Nomenclature: See separate page